Paleomagnetism of basic intrusions from the Brock Inlier, Northwest Territories, Canada[1]

J. K. PARK

Geomagnetic Laboratory, Earth Physics Branch, Department of Energy, Mines and Resources, Ottawa, Ont., Canada K1A 0Y3

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Basic intrusions of the Brock Inlier of the Northwest Territories are shown by means of paleomagnetism to be related to the Franklin diabases (specifically the Coronation suite) of the Canadian Shield, thus verifying previous correlations made through geological considerations and radiometric dating. A predominantly reverse (A_R) and occasionally normal (A_N), presumably magnetite, magnetic component were resolved in single specimens from five separate sills (five sites) using alternating field and subsequent thermal treatment in conjunction with vector diagrams. A presumably hematite component antiparallel to A_R was recovered at one site using converging great circles. The mean A direction is at 074°, −07° ($k = 15$) and the average of the site poles at 165°E, 02°S ($N = 5$, $K = 40$, $A_{95} = 12°$), close to that of the Coronation suite. Two specimens from a single dyke site yielded a normal A_N-type pole similar to that of the Franklin dykes to the east of the Coronation suite.

Il est démontré au moyen d'une étude paléomagnétique que les intrusions basiques de la fenêtre Brock dans les Territoires du Nord-Ouest sont reliées aux diabases Franklin (particulièrement la séquence Coronation) du Bouclier canadien, confirmant ainsi les corrélations antérieurement réalisées à partir de considérations géologiques et de datations radiométriques. Une composante magnétique inverse (A_I) prédominante et une composante normale (A_N) occasionnelle, supposément dues à de la magnétite, ont été observées sur des spécimens individuels prélevés dans cinq filons-couches différents (cinq sites) en appliquant la méthode du champ alternatif suivie par traitement thermique en conjonction avec des diagrammes de vecteurs. Une composante antiparallèle à A_I, résultant probablement de l'hématite, a été détectée à un site à l'aide des grands cercles convergents. La direction moyenne de A est à 074°, −07° ($k = 15$) et la moyenne des pôles des sites est à 165°E, 02°S ($N = 5$, $K = 40$, $A_{95} = 12°$), valeur voisine de celle observée dans la séquence Coronation. Deux échantillons provenant du même site d'un dike ont donné un pôle de type normal A_N semblable à celui des dikes Franklin à l'est de la séquence Coronation.

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Introduction

The northern Canadian Shield is host to an extensive mafic suite of intrusive rocks and their probable extrusive equivalents, informally called the Franklin diabases (Fahrig *et al.* 1971). It has been suggested that they represent a late expression of the same tensional regime that gave rise to the Hayhook Orogeny of the northern Cordillera during the Hadrynian (Young *et al.* 1979). These widespread rocks, which range from Great Bear Lake to Baffin Island and northern Quebec, have been related together through geological and age studies, and through paleomagnetism (Fahrig *et al.* 1971; Robertson and Baragar 1972; Fahrig and Schwarz 1973; Park 1974; Palmer and Hayatsu 1975). North of Great Bear Lake intrusions of a similar nature occur in the Brock Inlier. It had been suggested through both geological and age studies that these too belonged to the Franklin diabases — in particular the Coronation suite (Young 1977). No correlation had been attempted through paleomagnetism, and it was partly with this in mind, following a suggestion by J. D. Aitken of the Geological Survey of Canada, that a paleomagnetic study of these rocks was undertaken. It was also hoped to determine whether they could be similarly correlated with the sills in the Tsezotene Formation of the Mackenzie Mountains, which at that time had also been sampled for paleomagnetic purposes. For logistical reasons the study was concentrated in the southern portion of the inlier (Fig. 1).

Gabbro (and diabase) sills and dykes occur throughout the Brock Inlier, ranging in composition from olivine to quartz gabbro. This unit (informally called P6, Cook and Aitken 1969) intrudes all strata of the Helikian (Hadrynian?) sedimentary group of the area (P1 to P5). The sediments have been assigned to the Shaler Group "because they lie on trend with the Group at its type locality on Victoria Island, and are similar in lithology and stratigraphic position" (Cook and Aitken 1969; see also Balkwill and Yorath 1970). All these rocks were folded and faulted at least as late as the Devonian. In the sampling area beds dip less than 6° to the north or northeast. Samples were collected from six sites: five from what appear to be different sills intruded within a stratigraphic thickness of some 300–400 m and one (site

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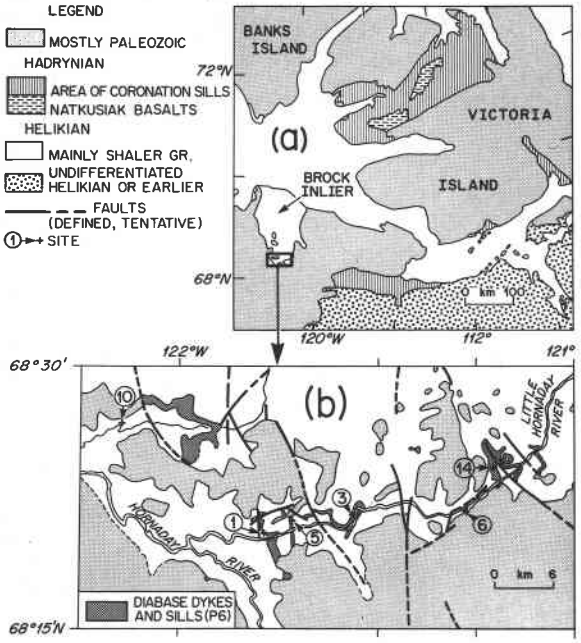


FIG. 1. General geological map (*a*) showing the sampling area (*b*).

5) from a 50 m thick dyke (Fig. 1*b*). One of the diabases designated as a sill (site 10) had not been previously mapped.

K–Ar age studies tend to confirm these intrusions as Franklin diabases. Two whole-rock ages (705 and 777 Ma) previously obtained from gabbro samples 50–100 km north of the sampling area (Wanless *et al.* 1965, p. 48) are similar to those obtained from the Coronation sills (five determinations, 604–718 Ma, average 641 Ma; Wanless *et al.* 1966, 1970). Fahrig *et al.* (1971) calculated an average K–Ar age of 647 Ma from the many determinations made on the Franklin diabases (including those from the Coronation sills), but preferred an age of 675 Ma based on probable argon loss. Sills within the Tsezotene Formation of the Mackenzie Mountains Supergroup have been dated by Rb–Sr at 766 ± 24 and 769 ± 27 Ma ($\lambda = 1.42 \times 10^{-11}$ year^{-1}; R. L. Armstrong, G. H. Eisbacher, and P. D. Evans, manuscript in preparation; J. D. Aitken, personal communication, 1981), and are thus of similar age.

Methods

All sites except site 10 were drilled in the field, and two or three standard specimens cut from each of five samples per site (except site 10) for a total of 68 specimens. Experiments to analyze the natural remanent magnetization (NRM) of specimens involved alternating field (AF) treatment followed by thermal demagnetization. AF treatment was carried out on the apparatus described by Roy *et al.* (1973) in spaces of low residual field (usually <2 nT) (Roy *et al.* 1969), and thermal demagnetization in Schonstedt TSD-1 furnaces. All measurements were made on Schonstedt DSM-1 spinner magnetometers. Magnetizations were generally resolved by using vector diagrams (Zijderveld 1967), individual components being obtained by estimating straight lines through linear segments of the plots defined by four or more points. Some 50% of the specimens were eliminated from the study because their vector diagrams did not reveal any linear segments that would indicate components with distinctive unblocking temperature (T_{UB}) or resistive coercive force (rcf) spectra.

Results and analysis

Progressive AF treatment of the sill specimens revealed two magnetic components; subsequent thermal treatment confirmed one of these and indicated two more.

The first component indicated by AF treatment is a generally steep downward-directed magnetization (*C*), which was removed in fields of from 5 to 25 mT (Fig. 2*b*). Only six *C* directions were accurately determined, one being steeply upward. By reversing the upward component the average *C* direction is 026°, +87° (k = 10, α_{95} = 22°), very close to Earth's field direction

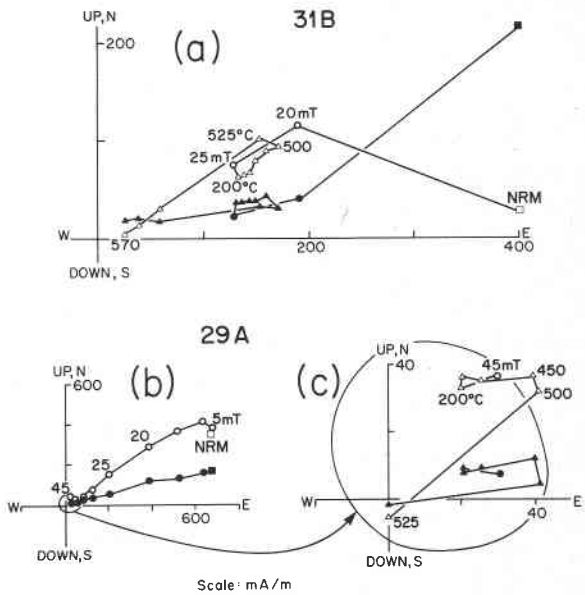


FIG. 2. Vector diagrams showing the effect of two-stage treatment (AF followed by thermal) on the remanent magnetization of specimens from a sill site (6). Circular symbols represent AF data and triangular symbols thermal. Closed symbols represent vectors plotted on the *XY* (N–E) horizontal plane and open symbols vectors plotted on the *ZY* vertical plane; both plotted after tilt correction.

(043°, +84°) in the sampling area. *C* was subsequently found by means of viscous remanent magnetization (VRM) acquisition experiments to be a VRM acquired in the recent Earth's field.

The second component is a shallowly inclined easterly or reverse magnetization (A_R), which was generally revealed by AF's up to between 40 and 75 mT (Fig. 2). Many specimens that yielded apparently valid reverse end points on stereographic projections under AF were found, on examination of the vector diagrams, to have mixed components in the end point range. The subsequent thermal treatment tended to resolve many of these mixed components, revealing a reverse component in general agreement with A_R and having T_{UB}'s in the approximate range of 525–570°C (Figs. 2, 3). Below about 525°C, five of the specimens exhibited a third component with a shallowly inclined normal westerly direction (A_N) (Fig. 2). Another specimen exhibited opposite behaviour, namely, an A_R component below 525°C and an A_N component above. Because A_R and A_N have complementary T_{UB} spectra in the temperature range to 570°C and are antiparallel within error, they presumably reside in magnetite.

The fourth magnetic component (B_N) was revealed above the T_{UB} range of A_R. It is well documented only at site 6, where its existence is confirmed in individual specimens by the presence of consecutive demagnetization directions that lie along great circles (Fig. 3). These circles do not coincide but instead have angular separations of up to 110° and, therefore, of course, *only* define an antiparallel direction to A_R. This component is likely present at other sites as well, but has probably been masked by the commonly observed acquisition of a spurious magnetization at some stage above 500°C. In site 6 this presumed spurious component is not evident until higher temperatures are reached, generally above 550°C, at which point vectors diverge from a great circle path and exhibit erratic intensities (Fig. 3). It is probable that B_N is carried by hematite.

Only two stable specimen directions were obtained from AF treatment of the dyke, and these are essentially antiparallel to the main A_R direction obtained from the sills.

Average *A* directions are given in Table 1 and individual specimen directions plotted in Fig. 4. All directions are quoted with reference to the bedding plane. No fold test was possible, but *A* is probably pre-folding and primary (see "Discussion"). Compared to the tilt-corrected direction of the sills (N = 5 sites; 074°, −07°; Table 1) the uncorrected direction is 074°, −04°. Thermal data (N = 18 specimens; 079°, −17°; α_{95} = 10°) are in reasonable agreement with AF data (N = 16 specimens; 074°, −05°; α_{95} = 12°), and are combined within sites, normal and reverse directions within specimens being treated as separate directions.

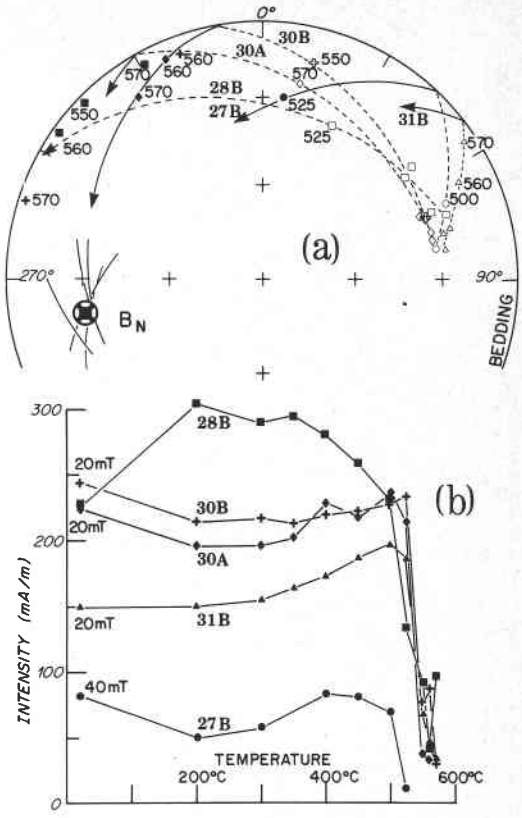


FIG. 3. Thermal demagnetization of the presumed magnetite component A_R at site 6 showing the great circle trajectories of the directions (*a*) and the intensity decay curves (*b*). Above about 550°C, data points tend to diverge considerably from their respective great circle paths, possibly owing to the acquisition of a spurious magnetization, and can be related to their respective great circles by noting the symbols used on the intensity decay curves.

Discussion

The facts that the *A* components are antiparallel within error and have largely non-overlapping T_{UB} ranges, that the diabases have not been metamorphosed, and that thorough experimental work has failed to reveal any other stable remanence except *B*, imply that the *A* magnetization is primary. It is therefore suggested that the *A* and *B* magnetizations were acquired during the initial cooling of the sills during a period of reversal of the Earth's magnetic field.

The results (all sites; pole: 165°E, 01°N) indicate that the intrusions of the Brock Inlier belong to the Franklin diabases (65 sites; pole: 163°E, 02°N; Palmer and Hayatsu 1975) and not to the sills in the Tsezotene Formation of the Mackenzie Mountains (pole: 139°E, 01°N; Park 1981). Any of these poles could be quoted as

TABLE 1. Summary of *A* directions and poles corrected for bedding

Si	Reverse directions (R)				Normal directions (N)				Average of N and R				Pole	
	Sp (Sa)	D (°) I (°)	R	k	Sp (Sa)	D (°) I (°)	R	k	Sp (Sa)	D (°) I (°)	R	k	α_{95}	Long. (°) Lat. (°)
1	4(2)	069 −16	3.91	33	—	— —	—	—	4(2)	069 −16	3.91	33	9	171E 00N
3	3(3)	079 −05	2.83	12	1(1)	238 +12	—	—	4(3)	074 −07	3.77	13	26	165E 02S
5	—	— —	—	—	2(2)	261 +34	2.00	—	2(2)	081 −34	2.00	—	—	164E 14N
6	10(5)	080 −30	9.91	98	2(1)	263 +54	2.00	—	12(5)	081 −34	11.76	45	7	164E 14N
10	4(3)	065 +19	3.92	39	—	— —	—	—	4(3)	065 +19	3.92	39	15	168E 18S
14	7(5)	081 +06	6.87	48	2(2)	267 +04	1.99	—	9(5)	082 +04	8.83	48	8	156E 05S
	*5**	*079 −05*	*4.75*	*16*	*3**	*255 +24*	*2.72*	*7*	*5**	*074 −07*	*4.73*	*15*	*21*	*165E 02S*
										Average of sill poles:	*4.90*	*40*	*12*	*165E 02S*
	28	*076 −10*	*25.90*	*13*	*5*	*259 +26*	*4.51*	*8*	*33*	*077 −12*	*30.24*	*12*	*8*	*163E 01N*

NOTES: Si is the site number; Sp is the number of specimens or sites (*) and Sa is the number of samples; D° and I° are the declination and inclination of the directions; R is the vector resultant; k is the precision; and α_{95}° is the error, the half angle of the 95% cone of confidence about the direction; the italicized data represent the grand averages of the sill data (i.e., excluding the dyke, site 5).

FIG. 4. All *A* specimen directions from both sills and dyke determined from vector diagrams.

an antipole, since there is no absolute reference scale of polarities for the Precambrian.

Palmer and Hayatsu (1975) have found by dividing all the results from the Franklin diabases (Fahrig *et al.* 1971; Robertson and Baragar 1972; Fahrig and Schwarz 1973; Park 1974; Palmer and Hayatsu 1975) into two groups that the average result from the central and eastern Arctic — from the Bathurst, Boothia, and Baffin Island dykes and sills (34 sites; pole: 165.5°E, 08.0°N; $k = 42, A_{95} = 3.8°$) — is significantly different from that of the "western" Arctic — from the Coronation sills, and Victoria Island sills and lavas (31 sites; pole: 161.1°E, 04.1°S; $k = 39$, $A_{95} = 4.2°$). They also noted that all the sites from the western Arctic have a single polarity (reverse), whereas those from the central and eastern Arctic are mixed. AF work on the sills as in previous studies confirms the reverse nature of the western Arctic results, though the thermal work (unlike the limited thermal work of previous studies: Robertson and Baragar 1972; Palmer and Hayatsu 1975) reveals the presence of normal components in some specimens. The combined AF and thermal data of the reverse directions give an average (five sites; 160°E, 02°S) within 3° of that quoted immediately above. No dykes had been sampled previously from the western Arctic, but the single dyke sampled for this study gave a *normal* direction like the Franklin dykes from the central Arctic (Fahrig *et al.* 1971). This direction yields a pole at 164°E, 14°N, similar to that cited for the central and eastern Arctic mixed direction (see above), though the significance of this agreement cannot be judged.

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